A disjoint ℓ_1 / positive c_0-quotient obstruction to the Bouras–Chen–Li L-weak compactness question

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Abstract

Chen and Li ask whether every almost Dunford–Pettis subset of a KB-space is L-weakly compact. We do not settle the question. We prove that any failure forces a rigid sequential obstruction in the dual: a disjoint sequence equivalent to the unit vector basis of ℓ_1. In a KB-space this sequence is weak* null and therefore induces a positive surjection onto c_0 which fails to be norm compact on the solid hull of the original set. That solid hull also contains a seminormalized disjoint, weakly-null Dunford–Pettis sequence. Thus a counterexample cannot arise unless this explicit positive disjoint quotient configuration exists.

1 Source question and status

For a Banach lattice E, a bounded set $A \subset E$ is *almost Dunford–Pettis* if every disjoint weakly-null sequence $(f_n) \subset E'$ is uniformly null on A. It is *L-weakly compact* if every disjoint sequence in its solid hull $\mathrm{sol}(A)$ is norm null.

Chen and Li prove that every almost Dunford–Pettis subset of a KB-space is relatively weakly compact, and leave the following stronger question open [1, p. 3]:

> Is every almost Dunford–Pettis set in a KB-space L-weakly compact?

They also identify it with coincidence of almost Dunford–Pettis and almost limited sets. El Fahri, Machrafi, and Moussa identify the same question with the assertion that every KB-space has the positive Dunford–Pettis relatively compact property (PDPrcP) [2, Question 2 and Theorem 3.15].

as desired. ☐

In a KB-space without the weak Dunford-Pettis property (e.g., ℓ_p, $1 < p < \infty$) almost Dunford-Pettis sets and relatively weakly compact sets can not coincide.

There is another topic included in Bouras' question:

(Q1) *Is every almost Dunford-Pettis set in a KB-space L-weakly compact?*

We are not able to answer it and have to leave it still open. However, we can make some comments on (Q1). In [3] Chen *et al.* introduced the class of almost limited sets in Banach lattices. A norm bounded subset A of a Banach lattice E is said to be an *almost limited set* if every disjoint, weak* null sequence (f_n) of E' converges uniformly to zero on A. Clearly, every almost limited set in E is an almost Dunford-Pettis set. Almost limited sets and L-weakly compact sets coincide in E if and only if the norm of E is order continuous ([3, Theorem 2.6(2)]). Therefore, (Q1) is the same as the following question:

(Q1$'$) *Do almost Dunford-Pettis sets and almost limited sets coincide in a KB-space?*

Let us recall that a Banach lattice E has the positive Schur property if, and only if, every relatively weakly compact subset of E is L-weakly compact. We know that every Banach lattice with the positive Schur property is a KB-space

2 The obstruction theorem

Call an operator $T : E \to c_0$ *disjoint* when its coordinate functionals $T'e_n \in E'$ are pairwise disjoint.

Theorem 1 (disjoint ℓ_1 obstruction)**.** *Let E be a Banach lattice. If some almost Dunford–Pettis set $A \subset E$ is not L-weakly compact, then E' contains a disjoint positive sequence equivalent to the unit vector basis of ℓ_1.*

If, in addition, E is a KB-space and $B = \mathrm{sol}(A)$, then there are:

(i) *a positive disjoint surjection $T : E \twoheadrightarrow c_0$ such that $T(B)$ is relatively weakly compact but not relatively norm compact; and*

(ii) *a seminormalized disjoint positive sequence $(x_n) \subset E$ which is weakly null and Dunford–Pettis.*

Proof intuition

Failure of L-weak compactness is detected by a bounded disjoint sequence of dual functionals which is not uniformly null on the solid hull. If a subsequence were weakly Cauchy, consecutive differences would be disjoint and weakly null, while the lattice seminorm of each difference would remain bounded away from zero. This contradicts the almost Dunford–Pettis property. Rosenthal's ℓ_1 theorem then forces an ℓ_1 subsequence. Order continuity in a KB-space turns that disjoint dual sequence into a weak* null sequence, hence into the coordinates of an operator to c_0; the ℓ_1 lower estimate makes the operator onto.

Proof. The solid hull $B = \mathrm{sol}(A)$ is again almost Dunford–Pettis [1, p. 2]. Put

$$\rho_B(f) = \sup_{x \in B} |f(x)| \qquad (f \in E').$$

Since B is solid, ρ_B is a lattice seminorm: $\rho_B(f) = \rho_B(|f|)$ and $0 \le f \le g$ imply $\rho_B(f) \le \rho_B(g)$.

The dual characterization of L-weak compactness [2, p. 3] supplies a norm-bounded disjoint sequence $(u_n) \subset E'$ for which $\rho_B(u_n)$ does not tend to zero. Passing to a subsequence and replacing u_n by $f_n = |u_n|$, we may assume that the f_n are positive, pairwise disjoint, and

$$\rho_B(f_n) \geq \delta > 0. \tag{1}$$

We claim that (f_n) has no weakly Cauchy subsequence. Otherwise write such a subsequence as (f_{n_j}) and put

$$h_k = f_{n_{2k}} - f_{n_{2k-1}}.$$

Weak Cauchyness gives $h_k \to 0$ weakly. The h_k are pairwise disjoint, because the f_n are positive and pairwise disjoint. But (1) gives

$$\rho_B(h_k) = \rho_B(|h_k|) = \rho_B(f_{n_{2k}} + f_{n_{2k-1}}) \geq \rho_B(f_{n_{2k}}) \geq \delta,$$

contradicting that B is almost Dunford–Pettis. Rosenthal's dichotomy [3] now gives a subsequence, relabelled (f_n), and a constant $c > 0$ such that for every finitely supported scalar sequence (b_n),

$$c \sum_n |b_n| \leq \left\| \sum_n b_n f_n \right\| \leq C \sum_n |b_n| \tag{2}$$

for some $C < \infty$. This proves the first assertion.

Assume now that E is a KB-space. Its norm is order continuous, hence every norm-bounded disjoint sequence in E' is weak* null. Therefore

$$T : E \longrightarrow c_0, \qquad Tx = (f_n(x))_{n=1}^{\infty}, \tag{3}$$

is a well-defined positive disjoint operator. Its adjoint satisfies $T'e_n = f_n$. Estimate (2) says exactly that $T' : \ell_1 \to E'$ is bounded below. By the closed-range/surjectivity theorem, T is onto c_0.

Chen–Li's theorem applies to the almost Dunford–Pettis set B, so B is relatively weakly compact. Yet (1) says

$$\sup_{y \in T(B)} |y_n| = \rho_B(f_n) \geq \delta.$$

Every relatively norm compact subset of c_0 has uniformly vanishing coordinate tails; hence $T(B)$ is not relatively norm compact.

Finally, because B is not L-weakly compact, it contains a disjoint positive sequence (x_n) with $\inf_n \|x_n\| > 0$. Lemma 3.7 of [2] applies to the relatively weakly compact almost Dunford–Pettis set B and shows that (x_n) is weakly null and Dunford–Pettis. □

Corollary 2. *The Chen–Li question has an affirmative answer for every KB-space E whose dual contains no disjoint sequence equivalent to the ℓ_1 basis. More generally, it has an affirmative answer whenever E admits no positive disjoint surjection onto c_0.*

Proof. Either failure would produce the prohibited object by Theorem 1. □

3 What remains for the full problem

The quotient in Theorem 1 is positive and disjoint, but it need not be a lattice homomorphism; its kernel need not be an ideal. Surjectivity therefore does not supply a disjoint positive lifting of the c_0 basis. If such a lifting were always available, its range would be a lattice copy of c_0, impossible in a KB-space, and the full conjecture would follow. This is exactly the point at which the direct upgrade stops. The theorem is thus a partial obstruction, not a claimed resolution of Q1 or of Wnuk's related positive-Schur problem.

References

[1] J. X. Chen and L. Li, *On a question of Bouras concerning weak compactness of almost Dunford–Pettis sets*, Bull. Aust. Math. Soc. 92 (2015), 111–114; arXiv:1609.06906.

[2] K. El Fahri, N. Machrafi, and M. Moussa, *Banach lattices with the positive Dunford–Pettis relatively compact property*, Extracta Math. 30 (2015), 161–179.

[3] H. P. Rosenthal, *A characterization of Banach spaces containing* ℓ_1, Proc. Natl. Acad. Sci. USA 71 (1974), 2411–2413.